

Macaulay2 Remark: It is OK to use Macaulay2 for difficult calculations. For example, for computing kernels of differentials. Also, as a warning, the `acyclicClosure` command only produces the correct output when the ring contains $\mathbb{Q}$.

Exercise 1.* Let $R = \mathbb{F}_2[[x, y]]/(x^2, xy)$. Let $R\langle Y \rangle$ denote the acyclic closure of k over R .

(a) Compute $R\langle Y_{\leq 5} \rangle$ without using the Macaulay2 package `DGAlgebras`.

(b) Find the first 5 deviations of R . Use Macaulay2 to verify your answer.

Exercise 2.* Let A be a graded-commutative dg R -algebra. Show that the natural map

$$A[x]\langle y \mid \partial y = x \rangle \longrightarrow A$$

is a homotopy equivalence of dg A -algebras.

Hint: First, a helpful reduction is to realize that there is an isomorphism of dg R -algebras

$$A \otimes_R R[x]\langle y \mid \partial y = x \rangle \cong A[x]\langle y \mid \partial y = x \rangle,$$

and so it suffices to show that $R[x]\langle y \mid \partial y = x \rangle \rightarrow R$ is a homotopy equivalence of dg R -algebras.

Exercise 3.!! Let x be a degree 2 variable and consider the graded algebras $A = \mathbb{F}_2[x]$ and $B = \mathbb{F}_2\langle x \rangle$. Find the acyclic closures of $\mathbb{F}_2$ over A and B , respectively.

Exercise 4. Let $R = Q/I$ with $(Q, \mathfrak{m}, k)$ a regular local ring and $I \subseteq \mathfrak{m}^2$ generated by a regular sequence. Use the quasiisomorphisms of local dg algebras (with residue field k)

$$K^R \xleftarrow{\sim} Q[X_1] \otimes_Q K^Q \xrightarrow{\sim} k[X_1]$$

and Problem Set 2 Exercise 6 to write down an explicit acyclic closure of k over R in terms of the following data¹: $\mathfrak{m}$ is generated by $t_1, \dots, t_n$, the ideal I is minimally generated the regular sequence $f_1, \dots, f_c$ and

$$f_i = \sum_{j=1}^n a_{ij} t_j \quad \text{for some } a_{ij} \in \mathfrak{m}.$$

Exercise 5. Does a similar explicit description for a minimal generating set for $H_1(K^R)$ hold for a general R , as the one given in Exercise 4? If so, explain. If not, give a counterexample.

Exercise 6.* Fix a field k . Let A be a graded k -algebra. Check that A^{Lie} is a graded Lie algebra over k , and that the functor $A \rightarrow A^{\text{Lie}}$ is right adjoint to $U: \mathbf{Lie}_k \rightarrow \mathbf{Alg}_k$. Is this adjoint pair an equivalence? Justify your answer completely.

Exercise 7. Show that the center $\mathfrak{g}$ of a graded Lie algebra L is an abelian Lie subalgebra of L . Furthermore, this defines a central polynomial subalgebra $U(\mathfrak{g})$ of $U(L)$.

¹This description is due to Tate.

Exercise 8.* Let k be a field and consider the graded k -vector space $L = k\{u_i, v_i \mid i \geq 0\}$ where $|u_i| = 2i$ and $|v_i| = 2i + 1$, so that $\dim_k(L_i) = 1$ for all $i \geq 1$.

(a) Check that L is a graded Lie algebra under the antisymmetric bracket given by

$$[v_i, v_j] = 0 \quad v_i^{[2]} = 0 \quad [u_i, u_j] = 0 \quad [u_i, v_j] = v_{i+j}.$$

(b) Find the central and nilpotent elements of L .

Exercise 9. For a graded Lie algebra L over k , show that $U(L)$ is finite dimensional k -space if and only if L is finite dimensional and $L = L_{\text{odd}}$.

Exercise 10. Show that $\text{Lie}(V)$ is isomorphic to the subalgebra of $T(V)^{\text{Lie}}$ generated by V .

Hint: Use the universal properties of the tensor algebra, the free Lie algebra, and the universal envelope to show that $U(\text{Lie}(V)) \cong T(V)$. The Poincaré–Birkhoff–Witt Theorem implies that $L \rightarrow U(L)^{\text{Lie}}$ is injective for any graded Lie algebra L .

Exercise 11. Show that if $U(L)$ is infinite dimensional, then $U(L)$ is torsionfree. That is to say, prove that $\text{tors}(U(L)) = 0$ where

$$\text{tors}(U(L)) := \{y \in U(L) \mid yx_{i_1} \cdots x_{i_n} = 0 \text{ for all } n \gg 0\}.$$

Exercise 12.* Let $Q = k[[x, y]]$ and $R = Q/(x^2, xy)$, with a k a field of characteristic zero. Compute the brackets $[a, b]$ for $a, b \in \pi^i(R)$ with $i = 1, 2$. Check your work with `Macaulay2`.

Exercise 13.* Assume R is a complete intersection ring with regular presentation $\hat{R} \cong Q/I$ satisfying that $I \subseteq \mathfrak{m}_Q^3$. Describe $\pi^*(R)$ as a graded Lie algebra over k , and use this description to calculate $\text{Ext}_R(k, k)$ as a graded k -algebra.

Exercise 14.* For $R = k[[x, y, z]]/(x^2 - yz, xz, y^2)$, use Problem Set 2 Exercise 4 to make a table for $\pi^*(R)$ recording the Lie brackets $[a, b]$ for $a, b \in \pi^i(R)$ with $i = 1, 2$.

Exercise 15. For

$$R = \frac{\mathbb{F}_7[[x, y, z, w]]}{(xy, z^2, zw, x^2z + w^3, y^2w + xw^2)}$$

use `Macaulay2` to calculate the Lie brackets $[a, b]$ for $a, b \in \pi^i(R)$ with $i = 1, 2$.